

Spatiotemporal Abnormal Behavior Detection Model Using ResNet50-LSTM



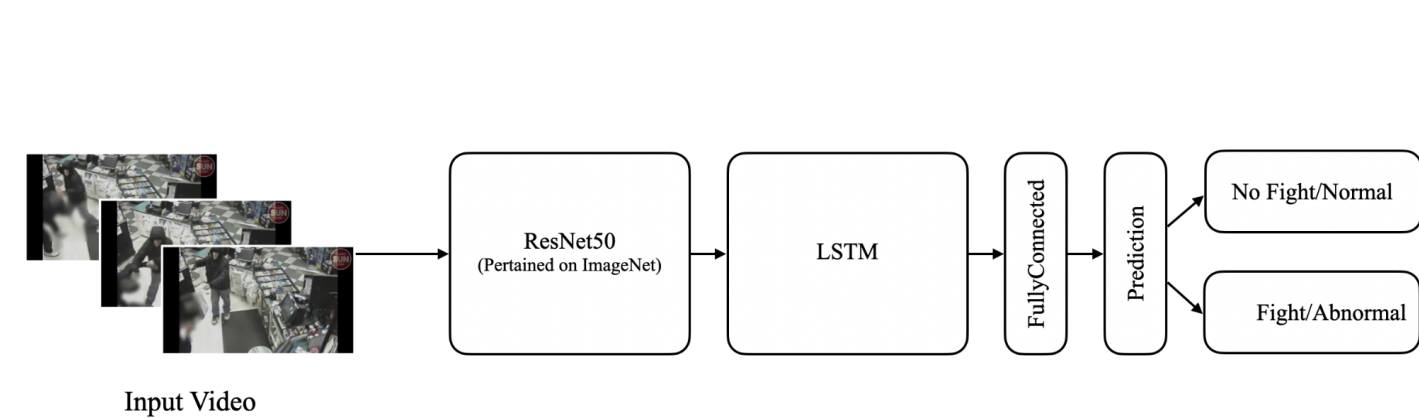
Student: Emmanuel Khamis Victor Loya | **Supervisor:** Dr. Felix Gonda

Problem & Motivation

- **Problem Context:** Manual CCTV monitoring is fatigue-prone and inconsistent; frame-based AI tools generate excessive false alarms.
- **Research Gap:** Existing models ignore temporal context and are mostly trained on synthesized, staged data rather than real footage.
- **Objective:** Develop and evaluate a hybrid ResNet50–LSTM model to detect fight-related abnormal behavior in real surveillance video.

Methodology & Dataset

System Architecture
Our system is an LSTM-based deep network using a **ResNet50** as a backbone for real-time abnormal behavior classification.



Dataset
We utilized the RWF-2000 data, which consist of 2,000 real-world CCTV clips, balanced Fight / Non-Fight (1,600 train / 400 test).



Evaluation & Key Results

Primary Metric
Our model achieved **72,5% test accuracy**. And **74% F1-score**.

Fight Recall: 80% of real fights correctly caught

Confusion Matrix			
		Pred: Fight	Pred: Non-Fight
Actual:	Fight	16	4
	Non-Fight	7	13

Significance & Conclusion

- **Contribution:** A validated ResNet50–LSTM spatiotemporal model for real-world fight detection, with a full evaluation framework.
- **Impact:** Usable as a decision-support alert tool for CCTV operators, reducing manual monitoring load.
- **Future Work:** Multi-class abnormal behavior detection, real-time edge deployment, and broader, more diverse training data.